Stellar giants

The vast majority of stars are too far away to have their radii measured directly. Size is usually estimated using a physical law that links radius, energy output, and surface temperature. Because many of the biggest stars pulsate, the resulting size accuracy is only about 10 percent. Theoretically, it has been estimated that giant stars become unstable if bigger than about 1,500 times the size of the Sun. The orbits of Earth and Jupiter are 215 and 1,120 the radius of the Sun, so all the stars in this list are bigger than Jupiter's orbit

LARGEST KNOWN STARS (BY RADIUS)

Name	Estimated radius (1=Radius of the Sun)	Type
UY Scuti	1,700	Red supergiant
NML Cygni	1,640	Red hypergiant
WOH G64	1,540	Red hypergiant
RW Cephei	1,535	Orange hypergiant
Westerlund 1-26	1,530	Red supergiant
V354 Cephei	1,520	Red supergiant
VX Sagittarii	1,520	Red hypergiant
VY Canis Majoris	1,420	Red hypergiant
KY Cygni	1,420	Red hypergiant
AH Scorpii	1,410	Red supergiant

Nearby stars and star groups

Over 90 percent of the Sun's neighbors are main-sequence stars, and 50 percent are in binary or triple groups. Typically, the average spacing is about seven light-years. It would take spacecraft such Voyager 1 about 100,000 years to travel this far.

Proxima Centauri will remain the closest star to the Sun for the next 25,000 years, after which Alpha Centauri takes over. This list will slowly change as the Sun travels around its Galactic orbit every 225,000,000 years.

CLOSEST STARS AND GROUPS

Name	Group	Component stars	Apparent magnitude	Absolute magnitude	Distance from Earth (in light-years)	Type
Sun	Single		-26.78	4.82	0.000016	Yellow main-sequence
Alpha Centauri	Triple	Proxima Alpha Centauri A Alpha Centauri B	11.09 0.01 1.34	15.53 4.38 5.71	4.2 4.4 4.4	Red main-sequence Yellow main-sequence Orange main-sequence
Barnard's Star	Single		9.53	13.22	5.9	Red main-sequence
Wolf 359	Single		13.44	16.55	7.8	Red main-sequence
Lalande 21185	Single		7.47	10.44	8.3	Red main-sequence
Sirius	Double	Alpha Canis Majoris A Alpha Canis Majoris B	-1.43 8.44	1.47 11.34	8.6 8.6	Blue-white main-sequence White dwarf
Luyten 726-8	Double	BL Ceti UV Ceti	12.54 12.99	15.40 15.85	8.7 8.7	Red main-sequence Red main-sequence
Ross 154	Single		10.43	13.07	9.7	Red main-sequence
Ross 248	Single		12.29	14.79	10.3	Red main-sequence
Epsilon Eridani	Single		3.73	6.19	10.5	Orange main-sequence
Lacaille 9352	Single		7.34	9.75	10.7	Red main-sequence
Ross 128	Single		11.13	13.51	10.9	Red main-sequence
EZ Aquarii	Triple	EZ Aquarii A EZ Aquarii B EZ Aquarii C	13.33 13.27 14.03	15.64 15.58 16.34	11.3 11.3 11.3	Red main-sequence Red main-sequence Red main-sequence
Procyon	Double	Alpha Canis Minoris A Alpha Canis Minoris B	2.66 12.98	2.66 12.98	11.4 11.4	White main-sequence White dwarf
61 Cygni	Double	61 Cygni A 61 Cygni B	7.49 8.31	7.49 8.31	11.4 11.4	Orange main-sequence Orange main-sequence